

Birla Institute of Technology & Science, Pilani, Rajasthan
First Semester 2025-2026
Tutorial-8

Course: ECE/EEE F311 Communication Systems
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Objectives

- M-ary, Random Variables

Tutorial Problems

1. Binary data is transmitted over a certain channel at a rate R_b bit/s. To reduce the transmission bandwidth, it is decided to use 8-ary PAM signaling to transmit this data.
 - (a) By what factor is the bandwidth reduced?
 - (b) By what factor is the transmitted power increased, assuming minimum separation between pulse amplitudes to be the same in both cases?
2. Solve the following:
 - (a) If the PDF of random variable X is $f_X(x) = e^{-x}$, $x > 0$, Find $P(X > 2)$.
 - (b) If the PDF of random variable X is Gaussian distributed with mean 0 and variance 1, Find $P(X > 2)$.
3. For the PDF

$$f_X(x) = \frac{1}{2\sqrt{2\pi}} e^{-x^2/32}$$

find $P(1 < x \leq 2)$

4. The noise voltage in an electronic circuit can be modeled as a Gaussian random variable $N(0, 10^{-8})$. Find the probability that the value of noise exceeds 10^{-4} .

Solution

1. Appended (BP Lathi 7.7-3):

7.7-3 (a) Each 8-ary pulse carries $\log_2 8 = 3$ bits of information. Hence, the bandwidth is reduced by a factor of 3.

(b) The amplitudes of the 8 pulses used in this 8-ary scheme are $\pm A/2$, $\pm 3A/2$, $\pm 5A/2$, and $\pm 7A/2$. Consider binary case using pulses $\pm A/2$. Let the energy of each of these pulses (of amplitude $\pm A/2$) be E_b . The power of this binary case is

$$P_{\text{binary}} = E_b R_b$$

Because the pulse energy is proportional to the square of the amplitude, the energy of a pulse $\pm \frac{kA}{2}$ is $k^2 E_b$. Hence, the average energy of the 8 pulses in the 8-ary case above is

$$E_{av} = \frac{E_b \left[2(\pm 1)^2 + 2\left(\pm \frac{3}{2}\right)^2 + 2\left(\pm \frac{5}{2}\right)^2 + 2\left(\pm \frac{7}{2}\right)^2 \right]}{8} = 21E_b$$

Hence,

$$P_{8\text{-ary}} = E_{av} \times \text{pulse rate} = 21E_b \times \frac{R_b}{3} = 7E_b R_b.$$

Therefore,

$$P_{8\text{-ary}} = 7P_{\text{binary}}$$

2. (a) $P(X > 2) = \int_2^\infty e^{-x} dx = e^{-2}.$

(b) $P(X > a) = Q(x/\sigma)$ to get $P(X > 2) = Q(2)$

3. $X \sim N(0, 4)$ with $\sigma = 2$; $P(1 < X \leq 2) = \int_1^2 f_X(x) dx = \int_1^\infty f_X(x) dx - \int_2^\infty f_X(x) dx = Q(1/\sigma) - Q(2/\sigma) = Q(1/2) - Q(2/2) = Q(0.5) - Q(1)$

4. Use $P(X > a) = Q(x/\sigma)$, where $X \sim N(0, \sigma^2)$. Here $\sigma = 10^{-4}$
 $P(X > 10^{-4}) = Q\left(\frac{10^{-4}}{10^{-4}}\right) = Q(1).$